TOPIC 5

Climate change

Topics

5.1 The natural and human causes of climate change
5.2 The impacts of climate change at a range of geographic scales
5.3 The responses to climate change

This topic looks at:

- the natural and human causes of climate change
- the impacts of and responses to climate change.

5.1 The natural and human causes of climate change

This chapter will explain:

★ the evidence for climate change
★ the factors that influence natural climate change
★ how human influence affects climate change.

The evidence for climate change

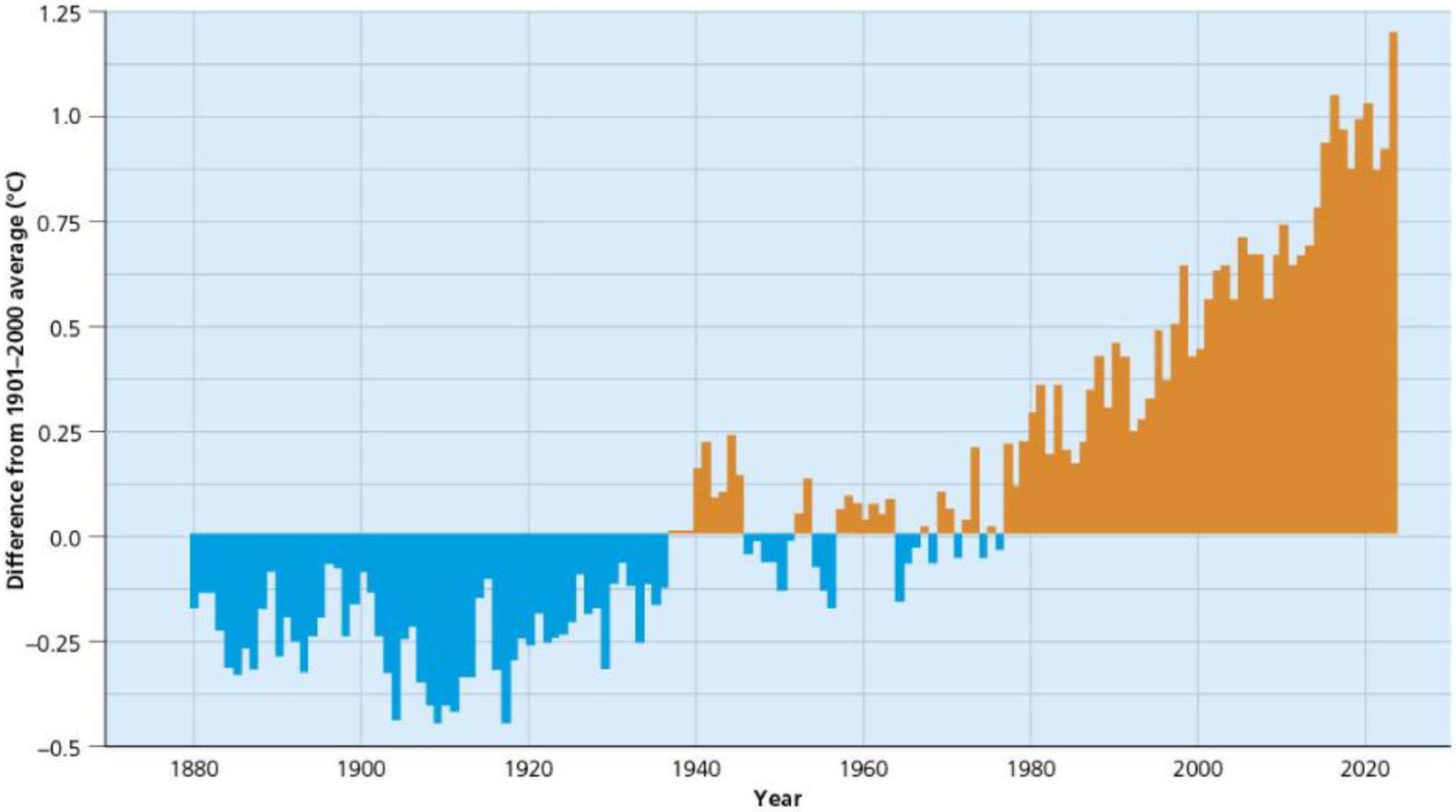
There is a range of evidence for climate change, including temperature data, ice cores, extent of sea ice and the records provided in historic writings and paintings.

Global temperature data

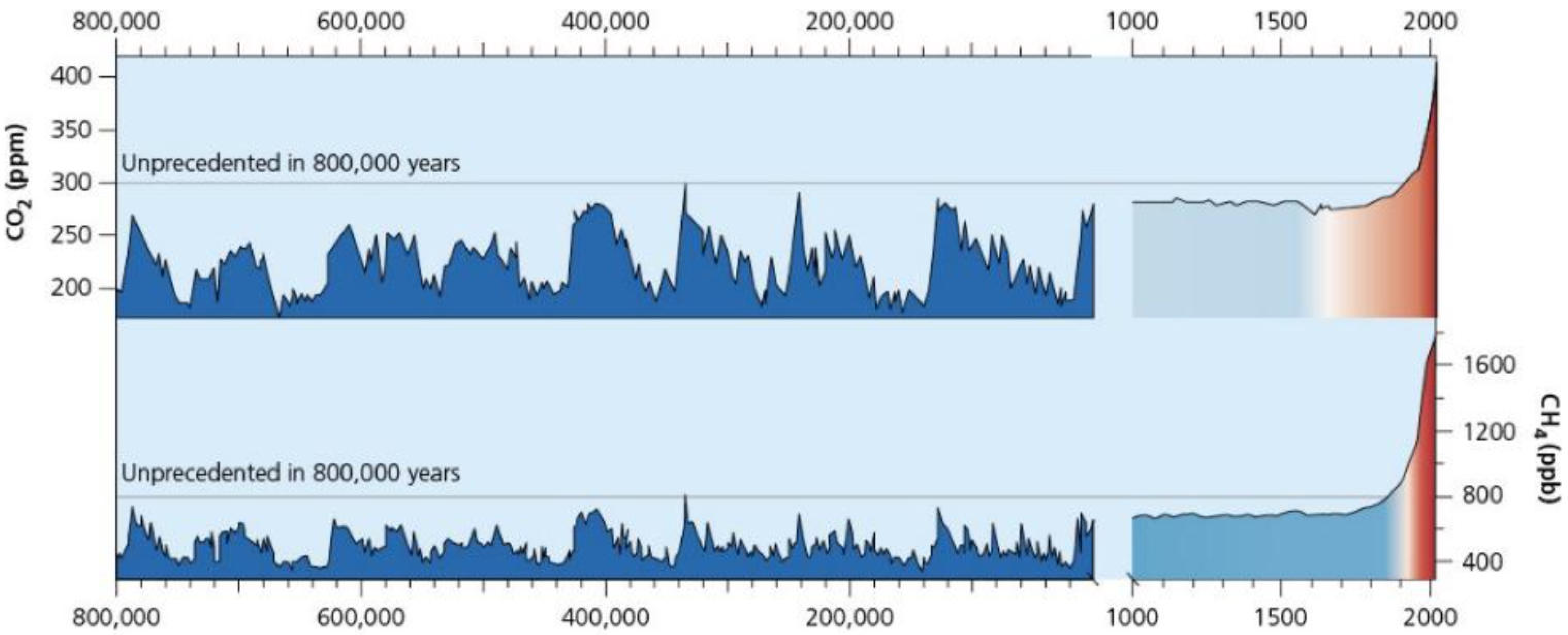
Average annual temperatures have increased by approximately 1.1°C since pre-industrial times. Figure 5.1 shows global average surface temperatures 1880–2023 compared with the average for the twentieth century. The upward trend in mean temperature is clear.

Ice cores

Ice core analysis shows that carbon dioxide (CO_2) levels, which had been steady for the previous 10,000 years at around 260–280 ppm, began to increase in the nineteenth century (Figure 5.2) and have now reached approximately 420 ppm, an increase of about 50 per cent in around 150 years. CO_2 is increasing at a rate of about 15 ppm every six years, while methane (CH_4) has doubled in the last 200 years.



▲ **Figure 5.1** Global average surface temperature 1880–2023 compared with the average for the twentieth century



▲ **Figure 5.2** Changes in atmospheric carbon dioxide (CO_2) and methane (CH_4) over the last 800,000 years, as recorded in ice cores and atmospheric sampling

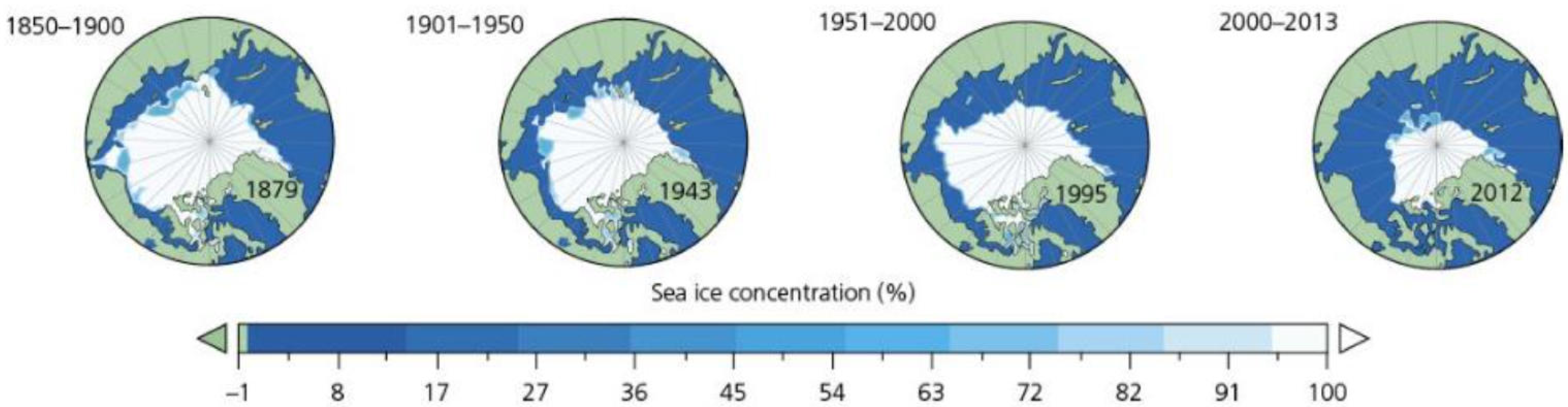
Interesting note

The history of climate change science is believed to have started with Eunice Newton Foote, an American scientist, inventor and women's rights campaigner. In 1856, she published a paper demonstrating the effects of CO_2 on temperature, in which she concluded: 'An atmosphere of that gas would give our Earth a higher temperature'.

Sea ice

Ice sheets, such as the West Antarctica ice sheet, are shrinking and thinning. Many of the world's glaciers are retreating and many smaller ones have disappeared. Snow cover in the Northern Hemisphere has decreased over the last 50 years and snow is melting earlier.

Levels of sea ice have been declining since the 1850s and the rate of this has accelerated. In each of the maps in Figure 5.3, the minimum amount of sea ice was towards the end of the period.



▲ **Figure 5.3** Sea ice cover maps for the annual minimum in September, for the periods 1850–1900, 1901–1950, 1951–2000 and 2001–2013

The decline of sea ice leads to two changes:

1. Sea levels rise.
2. There is a change in albedo as the light-coloured reflective surface of the sea ice is replaced by a darker ocean surface, which is less reflective and therefore absorbs more solar energy, reinforcing the heating process.

Sea levels have risen by about 16 cm since 1901, due to the expansion of water as it warms (the **steric effect**) and to melting ice caps and ice sheets. The rate at which sea levels rose between 2000 and 2020 was nearly double the rate of the twentieth century. In addition, the oceans are becoming more acidic as they take in some 20–30 per cent of total anthropogenic (human-made) carbon dioxide emissions. Since the Industrial Revolution, oceanic acidity has increased by 30 per cent.

Interesting note

The decline in snow and ice cover is having a major impact on species adapted to cold conditions, for example polar bears and Arctic hares. Many species may need to move to higher latitudes or altitudes, but the options for those already inhabiting high-altitude or high-latitude environments are limited. The same is true in oceans – some lower-latitude species such as tuna, cod and Alaska pollock are moving into higher-latitude oceans.

Historic writing and paintings

Some historic writings and paintings give clues about past climate. The first eight years of Charles Dickens' life (born 1812) had white Christmases (when snow falls on 25 December), and he refers to white Christmases in several of his books, including *The Pickwick Papers* (1836) and *A Christmas Carol* (1843).

Pieter Bruegel the Elder painted *The Hunters in the Snow* (1565) (Figure 5.4) following the harsh European winters of 1564 and 1565. Some people have wondered if *The Hunters* was a record of the start of the Little Ice Age.

We can conclude from these and other writings and art that there were colder periods during the past, and that these conditions were very different from the climate change that we are now experiencing.

▲ **Figure 5.4** *The Hunters in the Snow*, by Pieter Bruegel the Elder

Activities

1 Suggest how the painting *The Hunters in the Snow* by Pieter Bruegel the Elder and the writings of Charles Dickens suggest change in climate in Europe from the sixteenth century to the mid-nineteenth century to the present.

2 Describe the changes in Arctic sea ice between the periods 1850–1900 and 2000–13 shown in Figure 5.3 (page 111).

The natural causes of climate change

Orbital changes

The Quaternary Period refers to the most recent 2.6 million years of the Earth's history, up to and including the present day. The Quaternary has been characterised by frequent fluctuations of warming and cooling of the Earth, the growth and retreat of major ice sheets and ice caps, rises and falls of sea levels, the extinction of large mammals and the evolution of humans.

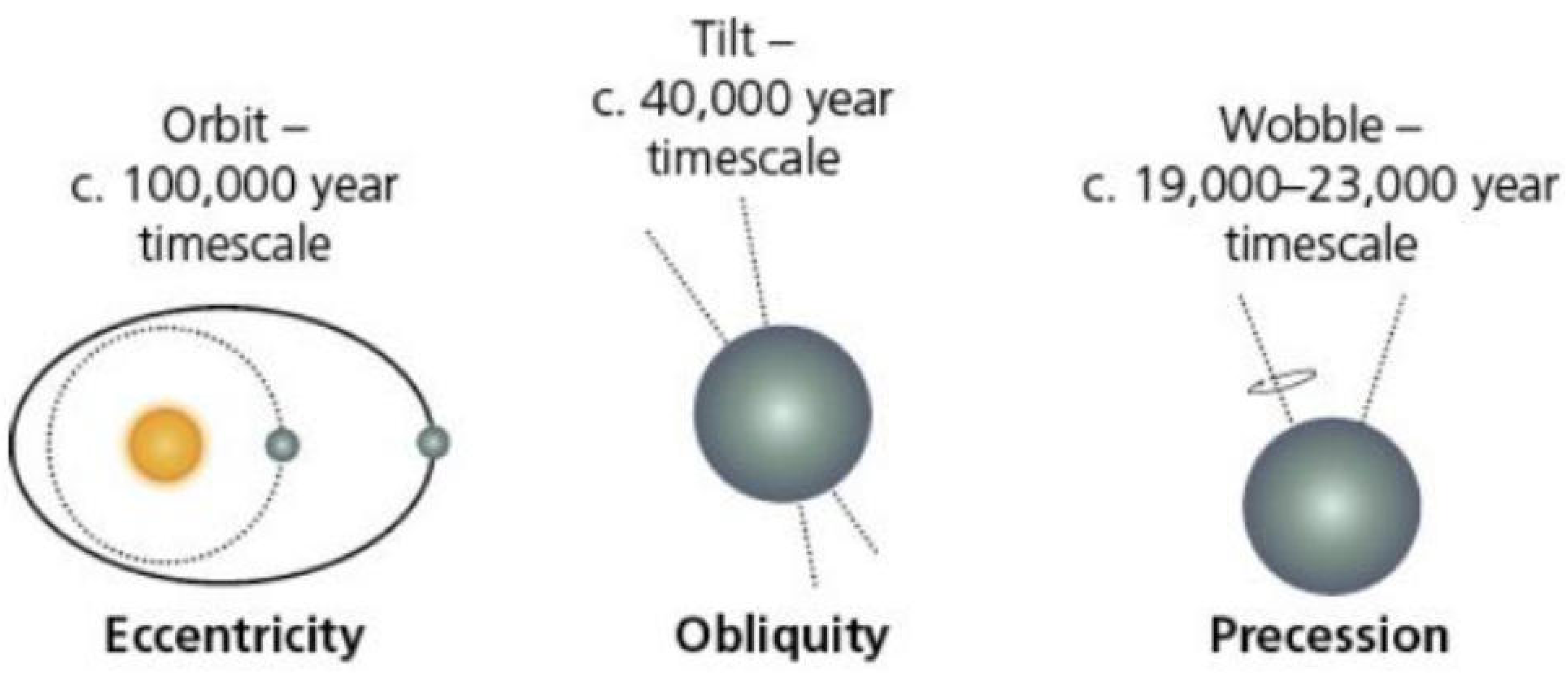
The Quaternary is a very short part of Earth's geological history. It is split into two parts, the Pleistocene Era and the Holocene Era:

- The **Pleistocene** refers to the geological era that lasted from about 2.6 million years ago to about 11,700 years ago. It was characterised by the growth and decline of ice sheets and ice caps.
- The **Holocene** refers to the most recent 11,700 years, when the Earth's climate has been relatively warm and the major ice sheets have retreated.

Nevertheless, the Earth is still in an ice age, as evidenced by the ice in Antarctica and Greenland. However, we are in a warm phase, known as an **interglacial period**. A cold phase, when ice grows, is referred to as a **glacial period**. Now, approximately 10 per cent of the Earth's surface is covered by ice. During the glacial periods, about 30 per cent of the Earth's surface is covered by ice.

Milankovitch cycles

One of the most important theories to help explain the natural causes of climate change in the Quaternary Period is that put forward by Milutin Milankovitch. The Milankovitch cycle theory (Figure 5.5) states that the amount of solar energy reaching the Earth varies with changes in the Earth's orbit, its tilt and its 'wobble' – that is, the direction of its rotation. The Earth's orbit varies over a timescale of about 100,000 years. When the Earth is further from the sun, it receives less energy. Also, when the tilt is greater, the seasons are longer. The 'wobble' determines which hemisphere – Northern or Southern – is facing the sun. This varies over a 19,000–23,000-year timescale.



▲ **Figure 5.5** The Milankovitch cycles

At the start of the Quaternary Period, the continents were largely in the positions that they are today. Since then, however, the Earth has wobbled in its rotation around the sun. These 'wobbles' have been partly responsible for the growth and decline of glaciations. By around 800,000 years ago a pattern had emerged: cold glacial phases that lasted up to 100,000 years, followed by a warmer interglacial period lasting about 10,000 to 15,000 years.

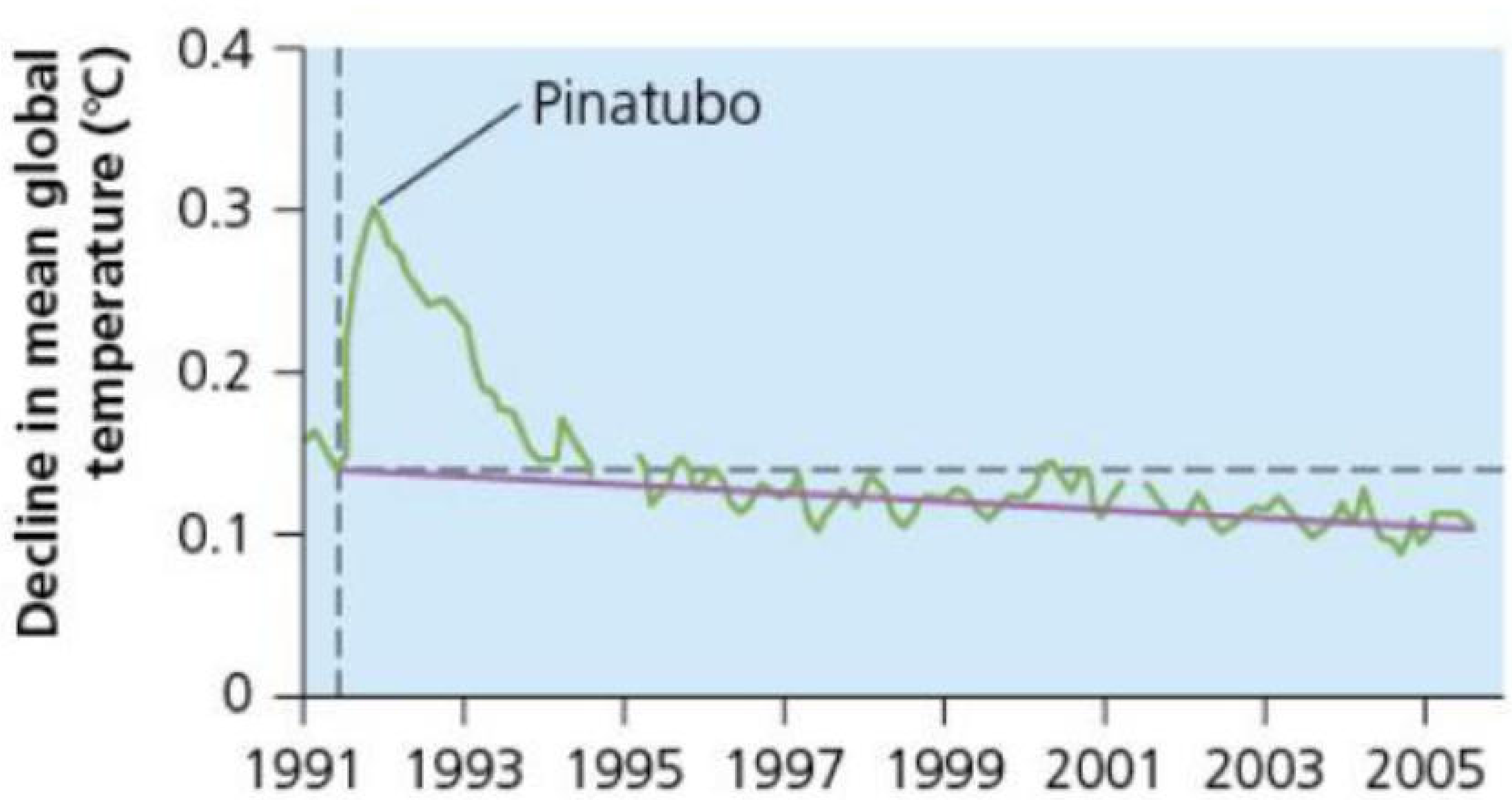
Sunspots

There is evidence that the sun has an 11-year cycle in which it brightens and dims – although the output only varies by around 0.1%. This change may have a very small effect on surface temperatures. Between 1880 and 1955, solar radiation gradually increased. However, since 1955 it has been gradually decreasing. So, although global temperatures have increased, due to global warming, solar output has decreased.

Volcanic activity

Other natural causes of climate change include volcanic eruptions (Figure 5.6). Those more likely to cause changes to climate are large eruptions, especially in tropical areas. The resulting atmospheric dust is also believed to block or reflect incoming solar energy, thereby leading to lower temperatures on Earth. Low latitudes receive a greater amount of solar energy than high latitudes, which is why volcanic eruptions in low latitudes have a greater impact on climate change – they block out a higher amount of solar radiation than high-latitude eruptions.

According to the United Nations' Intergovernmental Panel on Climate Change (IPCC), long- and short-term variations in solar activity play only a very small role in altering the Earth's climate.



▲ **Figure 5.6** The impact of the 1991 Mount Pinatubo eruption (Philippines) on temperature between 1991 and 2005: the purple line shows the global decline in sun-blocking aerosols following the eruption

Activities

1 Describe what happens to the Earth during:
  a the 100,000-year cycle
  b the 41,000-year cycle.
2 Explain how the eruption of Mount Pinatubo affected global climate.

The human influences on climate change

The **greenhouse effect** is the process by which certain gases (greenhouse gases) allow short-wave radiation to pass through the atmosphere but trap a proportion of the outgoing long-wave radiation from the Earth. This traps heat within the atmosphere like a greenhouse and leads to a warming of the atmosphere. It a natural process and vital for life on Earth.

The most common greenhouse gas is water vapour, which accounts for about 50 per cent of the natural greenhouse. However, the gases that account for human causes of climate change are carbon dioxide (CO_2), methane (CH_4) and chlorofluorocarbons (CFCs).

The enhanced greenhouse effect

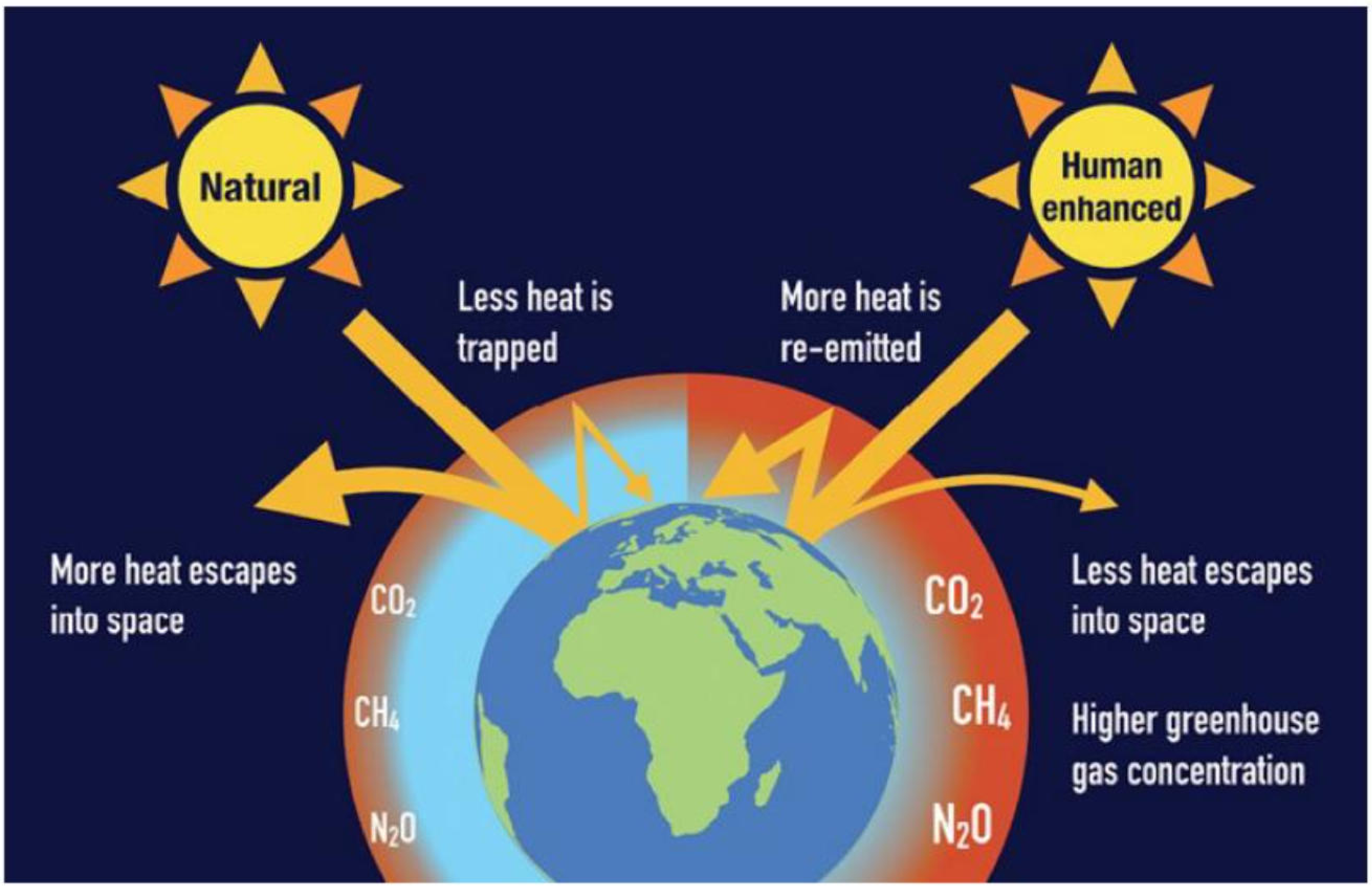
Atmospheric levels of carbon dioxide rose from around 315 parts per million (ppm) in 1950 to over 420 ppm by 2020, and they are predicted to rise to 600 ppm by 2050. This rise is due to issues such as burning **fossil fuels** (coal, oil and natural gas) and land-use changes to make way for agriculture, settlements and infrastructure. Deforestation reduces the store of carbon on the Earth (that is, in the trees) and carbon is released into the atmosphere when the trees are burned, adding to the amount of atmospheric carbon dioxide.

Agriculture is a major cause of climate change through processes including land-use change (deforestation, for example), methane emissions from cattle and rice fields, and the use of machinery in farming and in the transport of goods to market.

The **enhanced greenhouse effect** (EGHE) (Figures 5.7 and 5.8) is known by many terms, such as global climate change, climate crisis and global warming. As a result of the EGHE, temperatures have been rising since around 1850.

Activities

1 Identify three greenhouse gases.
2 Explain how greenhouse gases affect temperature on Earth.



▲ **Figure 5.7** The greenhouse effect and the enhanced greenhouse effect

▲ **Figure 5.8** Some impacts of global warming – sunbathing on glaciers and irrigation needed to play tennis

5.2 The impacts of climate change at a range of geographic scales

This chapter will explain the present-day and predicted future impacts of climate change:

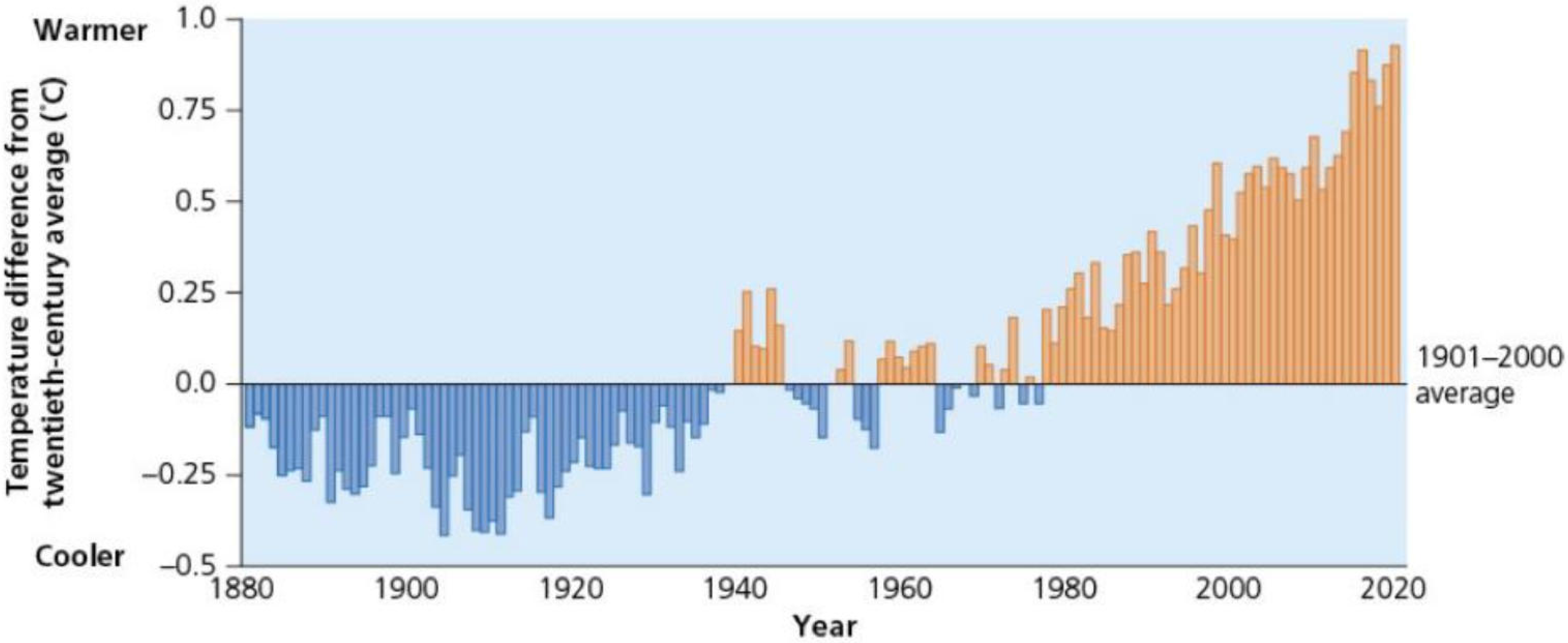
★ the changes to global temperature and weather patterns
★ the changes to sea levels
★ future predictions.

The changes to global temperature and weather patterns

Figure 5.9 shows the change in global surface temperatures between 1880 and 2020. Between 1880 and 1940, annual temperatures were below the 1880–2020 long-term average. It was coldest (about 0.4°C colder) between about 1900 and 1910. Temperatures generally fluctuated above and below the long-term average between 1940 and 1980. However, since 1980 temperatures have always been higher than the long-term average, reaching approximately 1.0°C hotter than the long-term average around 2015. It is predicted to be about 1.1–1.2°C above the long-term average in 2025.

According to a 2021 report by the IPCC (the United Nations body that assesses the science for climate change), the enhanced greenhouse effect should be seen as a 'code red' for humanity. It warns of increasingly extreme heatwaves, droughts, flooding and high temperatures over the next few decades. It also warns of an impending **tipping point**. It points out that:

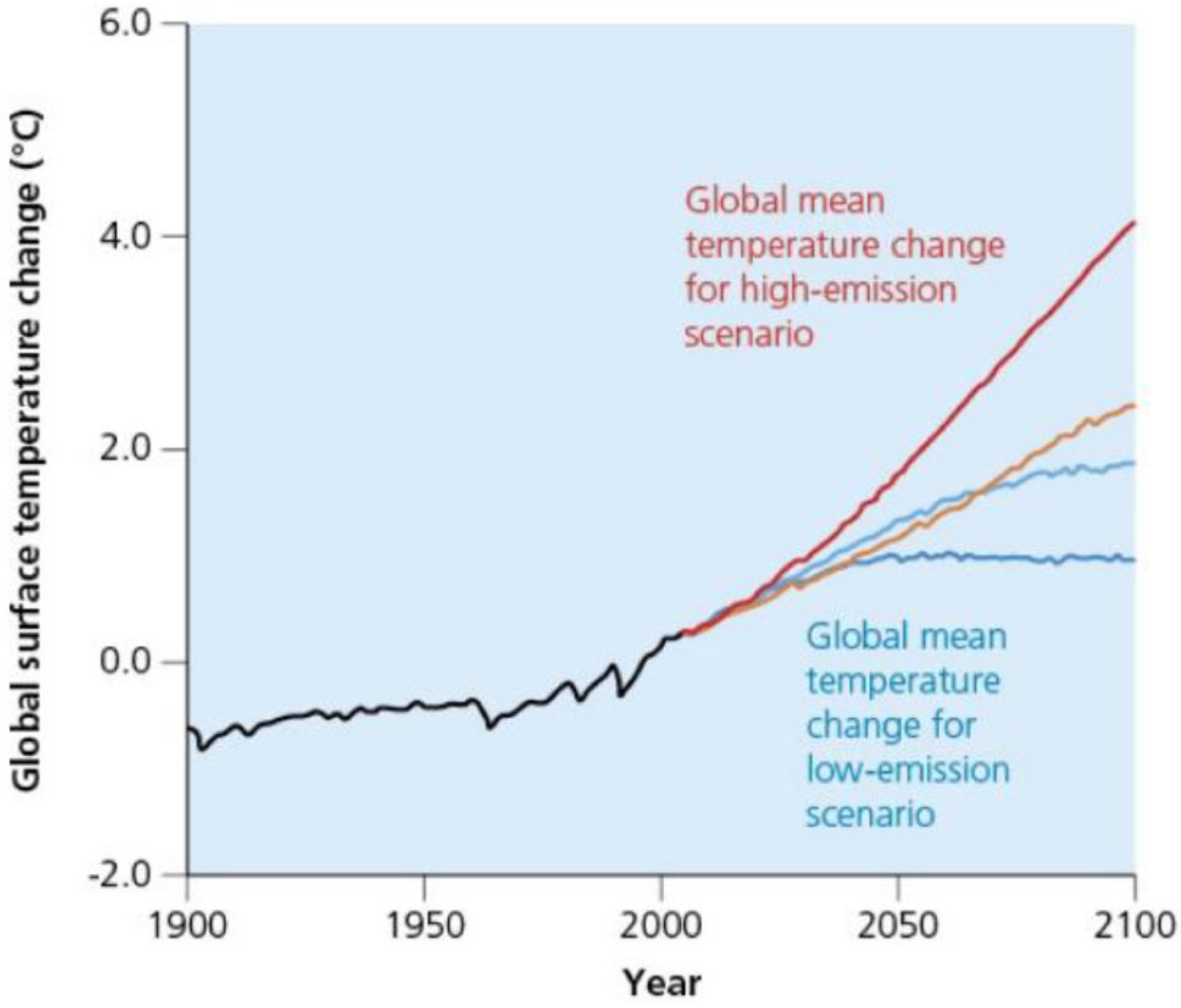
» since 1970, global surface temperatures have risen faster than in any other 50-year period over the last 2000 years
» global surface temperatures are about 1.1–1.2°C higher than in 1850–90
» the years 2017–24 were the warmest on record since 1850
» the rate of sea level rise has increased more than three-fold since 1971 – over 260 million people are at risk from rising sea levels and this could rise to 400 million people by 2100
» human influence is more than 90 per cent likely to be the cause of the global retreat of glaciers and the decrease of Arctic sea ice
» hot extremes have become more frequent and intense since the 1950s, for example there were droughts in the Amazon in 2005, 2010, 2015 and 2020, whereas cold extremes have become less frequent and less intense.



▲ **Figure 5.9** Annual global surface temperature change for land and ocean

The report also states that the enhanced greenhouse effect is much worse than previously thought, and that 40 per cent of the world's population are vulnerable to climate change. The report claims that climate extremes are going beyond humanity's and species' ability to cope. Between 2000 and 2020 there was a fifteen-fold increase in the number of deaths due to storms, floods and droughts, especially in emerging countries.

Extremes of weather are becoming more frequent and severe. The heatwave that killed over 70,000 people in Europe in 2003 was said to be a one-in-a-thousand-year event – but now it is said to be a one-in-ten-year event. In 2024, around 1300 pilgrims who made the journey to Mecca died due to heat exhaustion and dehydration. Temperatures there reached 50°C in the shade.

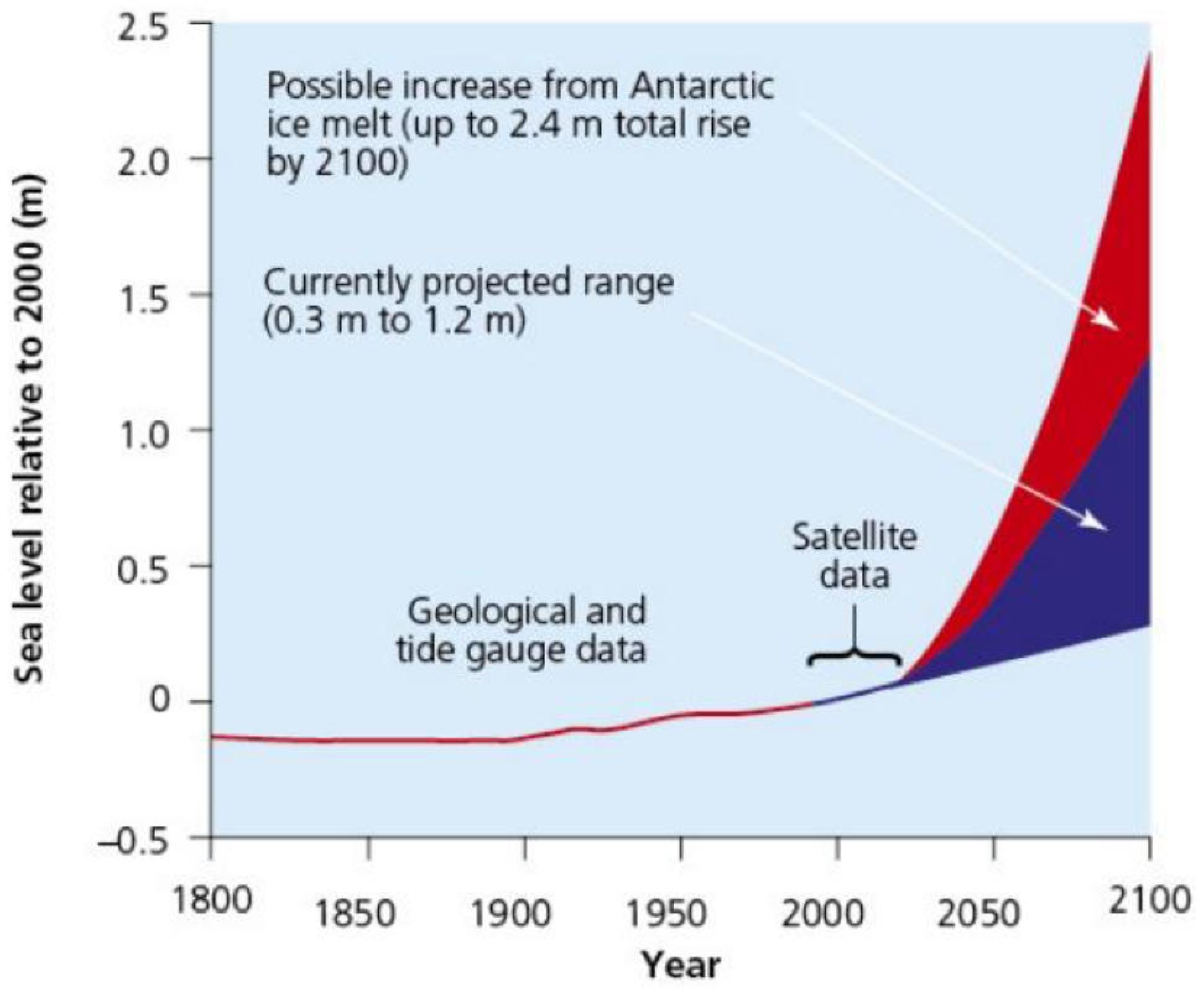


▲ **Figure 5.10** Predicted temperature changes 2000–2100 under high- and low-emissions scenarios

The changes to sea levels

Global mean sea level has risen about 21–24 cm since 1880. As already stated, the rising water level is mostly due to a combination of meltwater from glaciers and ice sheets and thermal expansion of sea water as it warms (the steric effect). Another, much smaller, contributor to sea level rise is the decline in the amount of water on land – in aquifers, lakes and reservoirs, rivers and soil moisture. This shift of liquid water from land to ocean is largely due to people depleting groundwater.

The global mean sea level in the ocean rose by 3.6 mm per year from 2006 to 2015, which was 2.5 times the average rate of 1.4 mm per year throughout most of the twentieth century. By 2100, global mean sea level is likely to rise by at least 0.3 m above 2000 levels (Figure 5.11). In the United States, about 30 per cent of the population lives in areas where sea level plays a role in flooding, shoreline erosion and hazards from storms.



▲ **Figure 5.11** Global mean sea level rise, 1800–2100 (projected)

Over 260 million people are at risk from rising sea levels, and by the year 2100 this could rise to over 400 million people. Over 60 per cent of these live in tropical regions. Many Pacific Ocean and Indian Ocean low-lying islands and many of the world's megacities are less than 10 m above sea level and coastal flooding is a major risk. In addition, food production in coastal areas is likely to be affected by saltwater intrusion. Extreme events that previously occurred once a century on average, are now predicted to occur annually. Up to 1 billion people are at risk from coastal-specific climate hazards such as tropical cyclones and storm surges.

Future predictions

Greenhouse gas concentrations in the atmosphere will continue to increase unless annual emissions decrease substantially. Increased concentrations are expected to raise average temperatures, alter patterns and amounts of precipitation, reduce ice and snow cover, increase the frequency, intensity and/or duration of extreme events, increase threats to human health, increase the acidity of oceans and cause ecosystems to shift their locations.

Precipitation is likely to increase, especially in tropical and high-latitude regions. The strength of the winds associated with tropical storms is likely to increase, although some scientists believe the storms will be more frequent but not necessarily stronger.

Sea ice and snow cover in the Northern Hemisphere has decreased since about 1970. Over the next century, these trends will continue.

Food production

Food production is likely to be affected by rising global temperatures. The decline in water resources may make it difficult for many farmers to continue the type of farming they currently practise. They may have to change the type of farming or the crops they grow, or they may even be forced out of farming altogether. Places in the Middle East, such as in Egypt, Dubai and Saudi Arabia, are likely to experience extreme heatwaves and farmlands may become desertified. In parts of Africa, especially Southern Africa, cereal crop productivity will be reduced as 30 per cent of the area that is used for maize production and 50 per cent of the area that is used for beans is likely to go out of production due to the enhanced greenhouse effect.

Activities

1 Compare the greenhouse effect and the enhanced greenhouse effect.

2 Using Figure 5.11 (page 116), explain why there are variations in the predicted sea level rise by 2100.

5.3 The responses to climate change

This chapter will explain:

★ the strategies used to manage the impacts of climate change
★ how successful mitigation and adaptation strategies and techniques have been in managing the impacts of climate change
★ detailed specific example: the impacts of, and responses to, climate change in Bangladesh.

The strategies used to manage climate change – national and international agreements

The major international agreements to manage climate change include the Paris Agreement, the Kyoto Protocol and the United Nations Framework Convention of Climate Change (UNFCCC).

The **UNFCCC** is the United Nations' (UN) process to negotiate an agreement to limit global climate change. To achieve this, it proposes to limit the amount of greenhouse gases in the atmosphere. It was signed in 1992 at the UN Conference on the Environment (the 'Earth Summit') in Rio de Janeiro. Its main objective is to stabilise greenhouse gases in the atmosphere. By 2022, the UNFCCC had 198 members ('Parties'), who meet annually at COP (Conference of the Parties) meetings.

The **Kyoto Protocol**, signed in 1997, became effective in 2005. In 2020 there were 192 member countries. The Kyoto Protocol aims to reduce global climate change by decreasing atmospheric greenhouse concentrations to a level that would prevent dangerous human interference. The Protocol had differentiated responsibilities, recognising that HICs were historically responsible for most greenhouse gas emissions. The Protocol's first commitment period was 2008–12. Although many HICs reduced their emissions, total global emissions increased from 1990 to 2010. A second agreement existed from 2012 to 2020. Some 37 countries had binding targets, including the EU (28 countries at that time), Australia and Ukraine. Some countries withdrew from the Protocol, including Canada and Russia.

The **Paris Agreement** (2015) is an international treaty on climate change, agreed by 196 countries. It is a separate agreement under the UNFCCC. The long-term goal of the Paris Agreement is to keep global climate change to less than 2°C above pre-industrial levels, and ideally to less than 1.5°C above pre-industrial levels. To achieve this, greenhouse gas emissions need to be reduced and net zero (that is, when emissions of greenhouse gases are balanced by removal of greenhouse gases from the atmosphere) reached by about 2050. The Paris Agreement has been criticised by some for not being strict enough. In 2017, the USA, under President Trump, announced that it would pull out of the Paris Agreement. It withdrew in 2020, but rejoined in 2021 under President Biden. The Paris Agreement does not cover emissions from international aviation or shipping.

Despite these agreements, greenhouse gases in the atmosphere continue to rise, as do global temperatures.

An evaluation of mitigation and adaptation strategies

Climate change **adaptation** refers to methods of learning to live with climate change, whereas **mitigation** refers to measures to prevent or limit global climate change (Figure 5.12). Adaptation is easier but less effective. Mitigation would be better in the long term, but is more difficult to achieve.

Adaptation includes:

» building higher flood walls
» using more air conditioning systems to cool living/work spaces
» ensuring availability of vaccines for diseases new to a region, such as malaria
» building flood shelters
» growing crops that are adapted to drought or heat.

Mitigation refers to attempts to reduce global climate change, including:

» burning less fossil fuel
» cutting down fewer trees
» planting more vegetation
» using sustainable transport
» conserving energy
» using renewable energy sources
» eating less red meat and dairy products.

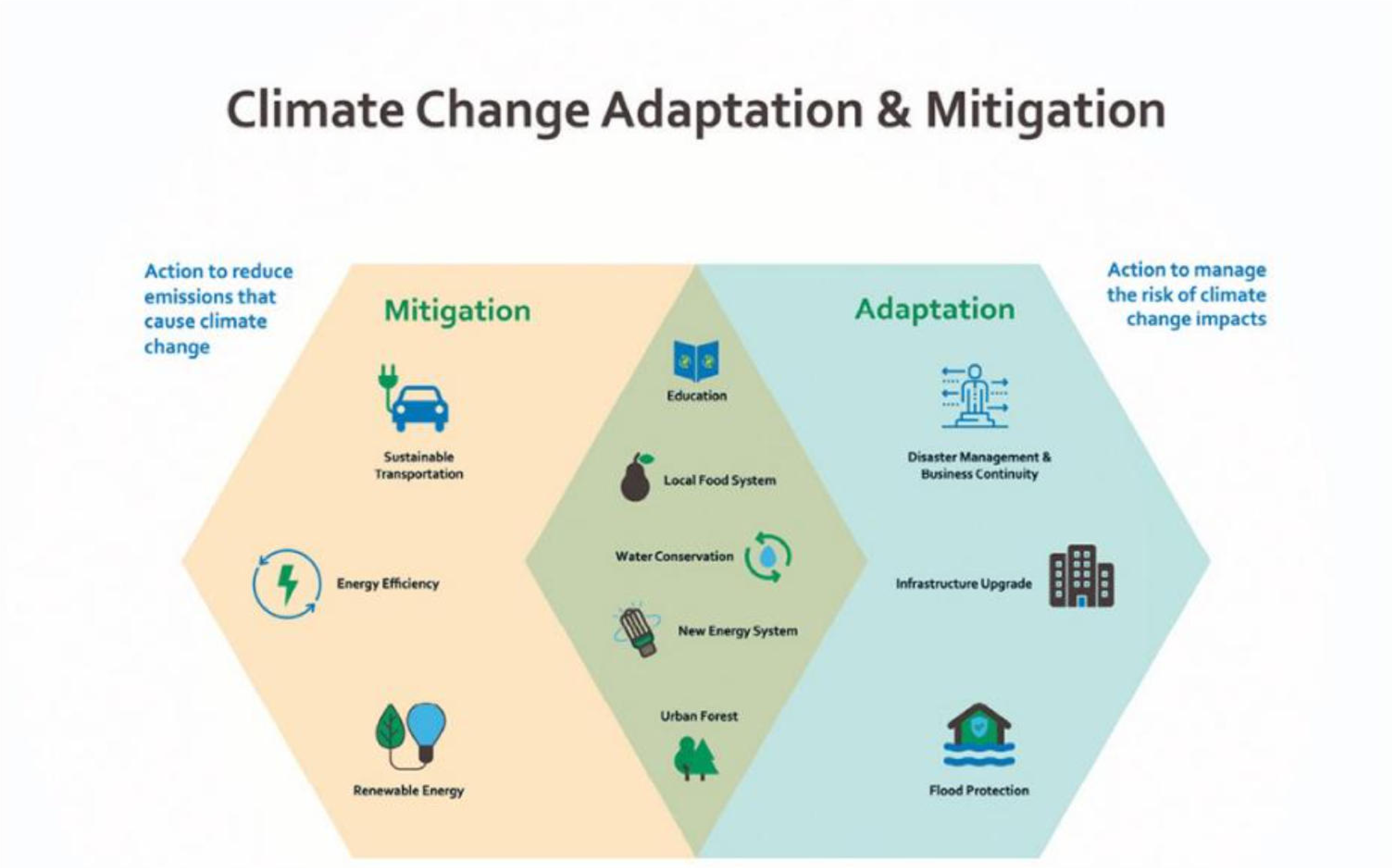
The disadvantages of mitigation include cost, for example the cost of developing nuclear power or building HEP schemes, and the difficulty of encouraging changes in people's behaviour, such as using less energy, using public transport, cycling or walking to work, college or school. The less mitigation that society achieves today, the more it will need to adapt in the future.

Adaptation may be easier, but it does not get rid of the problem. Adaptation will become progressively less effective if there is no mitigation, as global climate change will become more extreme over time. However, using clean energy sources, such as solar and wind power, will reduce the use of fossil fuels.

Other measures to reduce global climate change include insulating homes, switching off electricity sources when not in use, changing from a meat- and dairy-based diet to a vegetarian diet, reducing consumption of goods and using fewer plastics.

Activities

1 Suggest reasons why international agreements to manage the impact of climate change have been unsuccessful.
2 Explain how green infrastructure and water- and energy-conservation can be examples of adaptation and mitigation.



▲ **Figure 5.12** Adaptation and mitigation strategies for climate change

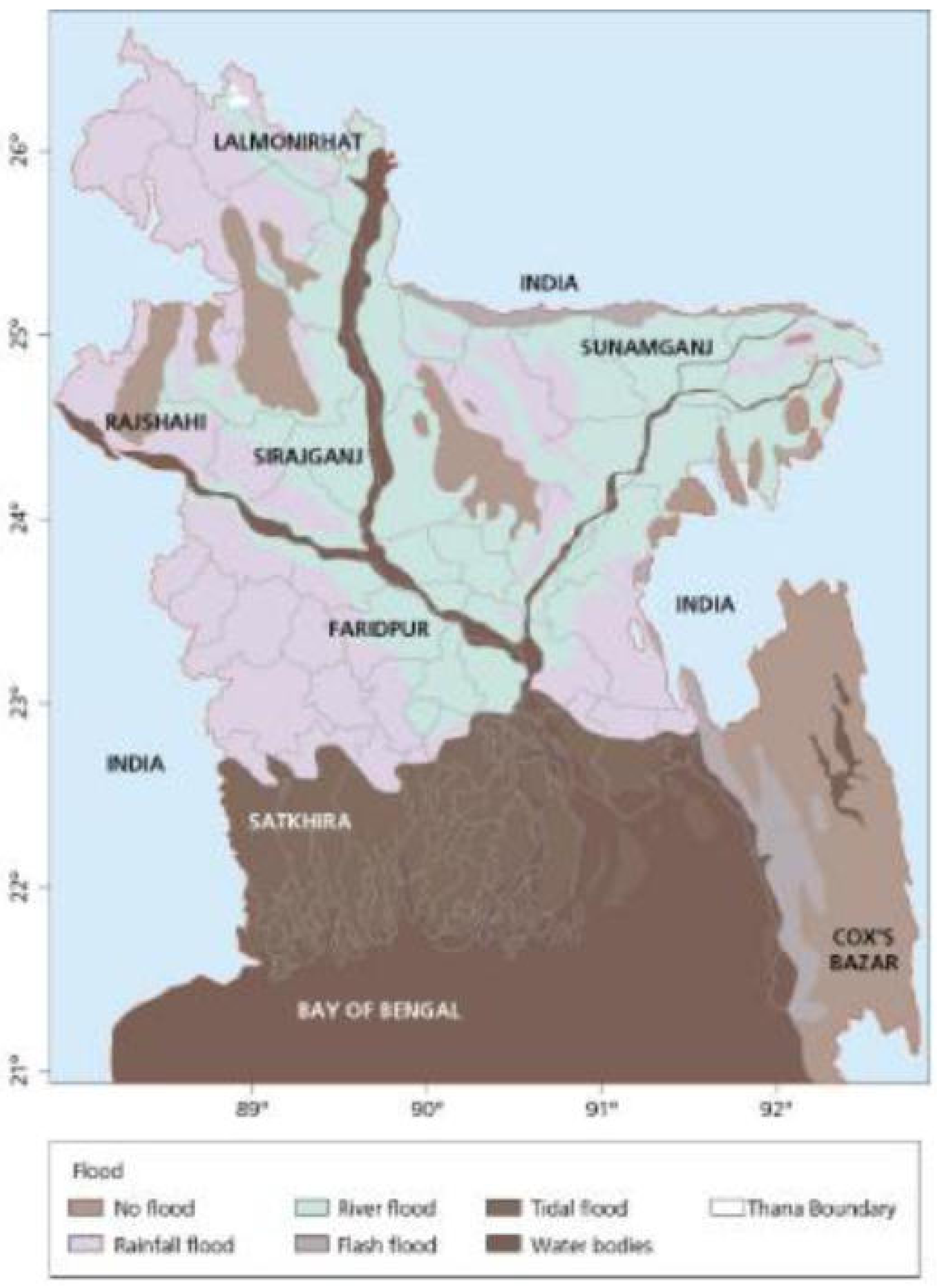
Detailed specific example

The impacts of, and responses to, climate change in Bangladesh

The impacts of climate change

Bangladesh accounts for less than 1 per cent of global greenhouse gas emissions, but it is one of the most vulnerable countries to climate devastation. This is known as a climate injustice.

Between 2000 and 2019 Bangladesh experienced economic loss of US$3.7 billion and 185 extreme weather events. Around 90 million Bangladeshis, over 50 per cent of the population, have a ‘high’ exposure to flooding. Two-thirds of Bangladesh is less than 5 m above sea level (Figure 5.13) and around one-third of the population live in coastal areas. A 50 cm rise in sea level could cause Bangladesh to lose 11 per cent of its land and up to 18 million people would be forced to leave their homes. A sea level rise of 2 m could displace up to 50 million people. Rising sea levels also cause salinisation, which is when salt accumulates in soil and severely reduces the soil’s fertility.



▲ **Figure 5.13** Flood risk in Bangladesh

Extreme events are becoming more frequent in Bangladesh. Up to 50 per cent of people living in Bangladesh's informal settlements are believed to be there due to flooding elsewhere. Dhaka, the capital, now has a population density of up to 47,500 people/km^2.

Tropical cyclones are becoming more intense. For example, in 2007 Cyclone Sidr brought winds of 240 km/h and killed over 3400 people. In 2009, Cyclone Aila led to the deaths of 200 people and left about 200,000 people homeless. Cyclone Amphan in 2020 was the strongest cyclone in Bangladesh's history. It led to the deaths of 128 people and caused US$14 billion worth of damage in Bangladesh and India.

The responses to climate change

To deal with such events, Bangladesh has developed a Climate Change Strategy Action Plan (BCCSAP). This was first produced in 2009, with five adaptation plans and one for mitigation. The Nationally Determined Contributions (NDCs) of 2021 planned:

- to reduce greenhouse gas emissions by >27 million tonnes of carbon dioxide equivalent (Mt CO_2e) (6.75 per cent) using Bangladesh's own resources by the end of 2030
- to reduce greenhouse gas emissions by >60 Mt CO_2e (>15 per cent) using international support by the end of 2030
- to reduce greenhouse gas emissions by around 90 Mt CO_2e (21.5 per cent) using combined resources by the end of 2030.

Strategies and techniques to manage climate change

Adaptation projects have included the building of embankments, over 4500 cyclone shelters and 300 flood shelters, and the excavation/re-excavation of canals.

Mitigation schemes have included the planting of millions of trees, including mangrove forests. As a result, emissions are expected to have been reduced by over 2 Mt CO_2e by 2025.

Bangladesh is pursuing a low carbon future with greater emphasis on renewable energy and increased energy efficiency. Farming methods used to adapt to the changing climate include:

- early-harvest, rapid-developing varieties, for example rice that develops in 100–20 days (traditional varieties require 140–50 days)
- drought-resistant early varieties
- salt-tolerant rice varieties
- flooding-tolerant crop varieties
- stress-tolerant crop varieties.

Bangladesh joined the Kyoto Protocol in 2013 and signed the Paris Agreement in 2016.

Activities

1 Describe the distribution and type of flood risk in Bangladesh, as shown in Figure 5.13 (page 120).
2 Distinguish between adaptation and mitigation strategies that have been used in Bangladesh.

Practice questions

1 Figure 5.14 shows annual global surface temperature change for land and ocean, compared to the 1901–2000 average.
   a Estimate the largest negative temperature difference from the twentieth-century average. [1]
   b Estimate the largest positive temperature difference from the twentieth-century average. [1]
   c Identify the decades when temperatures were lowest. [1]

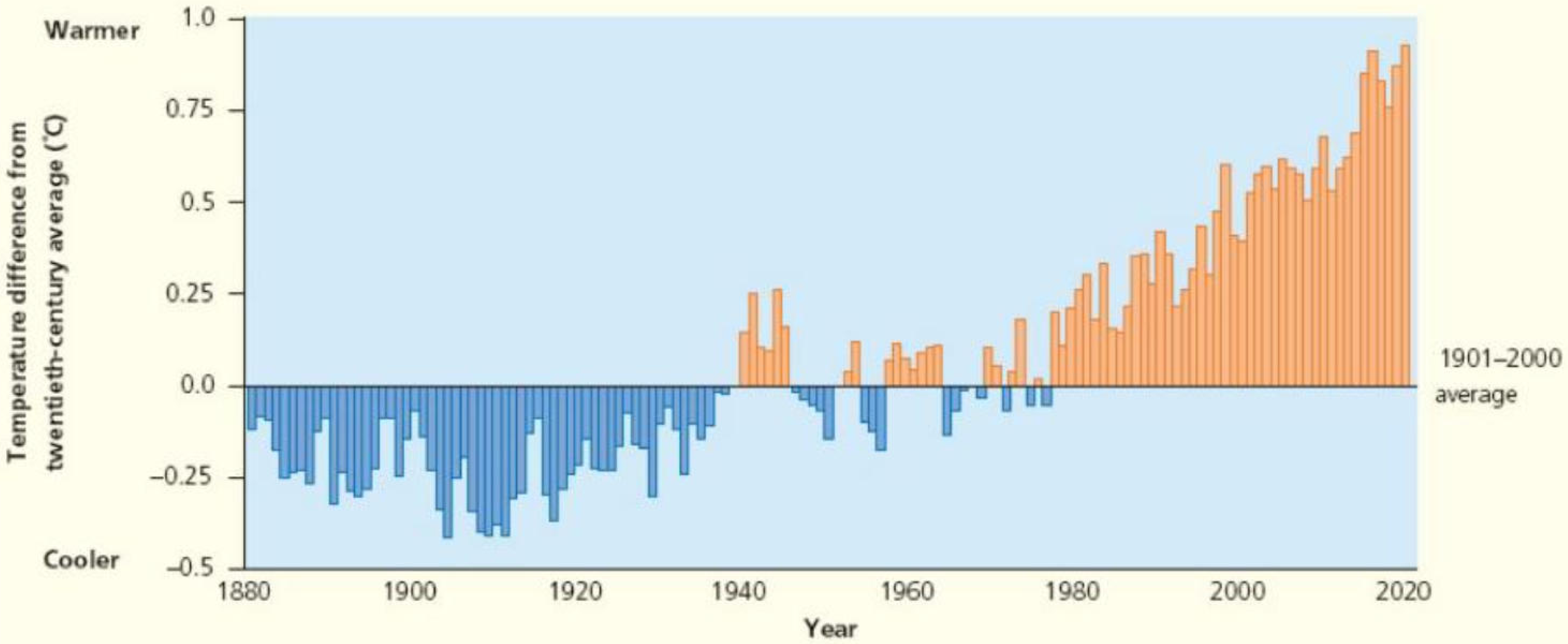
d Identify the decade when temperatures were highest. [1]

e Estimate the increase in temperature between the decades with the lowest temperatures and the decade with the highest temperatures. [2]

2 Distinguish between the greenhouse effect and global warming. [4]

3 **a** Define climate change mitigation and climate change adaptation. [2]

b Evaluate techniques of climate change mitigation and adaptation. [7]



▲ **Figure 5.14** Annual global surface temperature change for land and ocean, compared to the 1901–2000 average